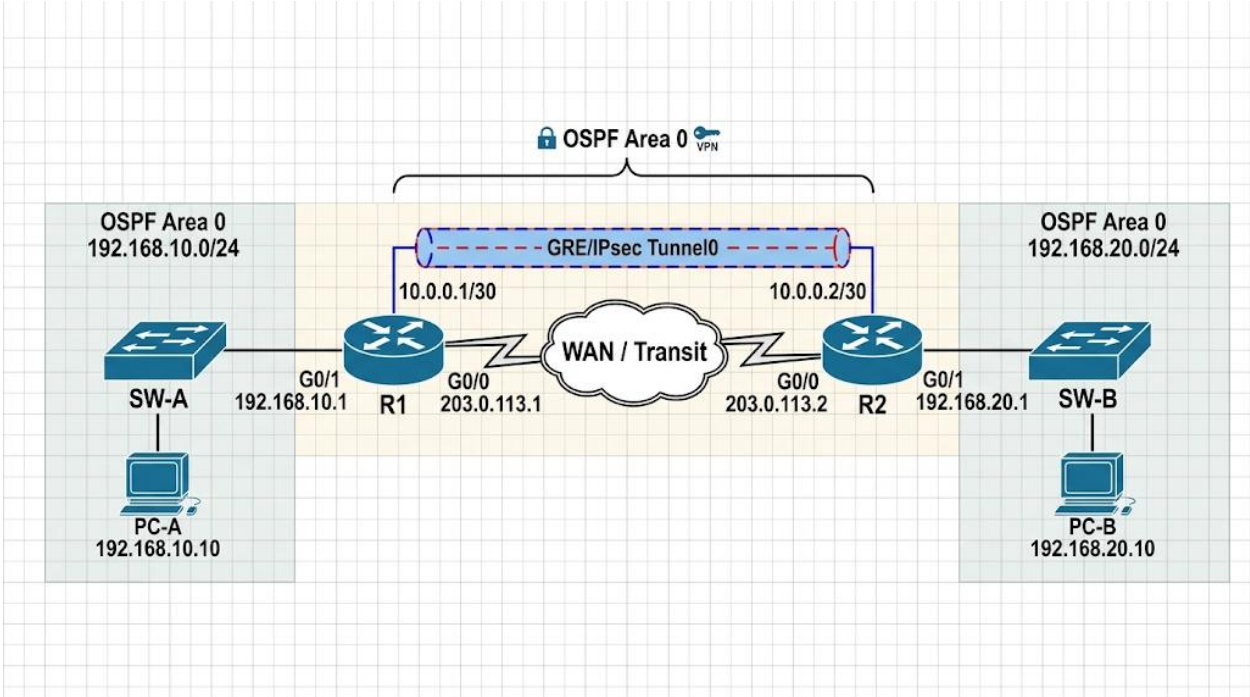
OSPF running over a GRE over IPsec VPN.



Here is the topology diagram and addressing scheme for the OSPF over GRE over IPsec lab.

Interface and Addressing Table

Device	Interface	IP Address	Subnet Mask	Role	OSPF Status
R1	GigabitEthernet0/1	192.168.10.1	255.255.255.0	LAN A Gateway	Advertised in Area 0
R1	GigabitEthernet0/0	203.0.113.1	255.255.255.252	WAN Link (Underlay)	Excluded from OSPF

Device	Interface	IP Address	Subnet Mask	Role	OSPF Status
R1	Tunnel0	10.0.0.1	255.255.255.252	GRE Encrypted Link	Advertised in Area 0
R2	GigabitEthernet0/1	192.168.20.1	255.255.255.0	LAN B Gateway	Advertised in Area 0
R2	GigabitEthernet0/0	203.0.113.2	255.255.255.252	WAN Link (Underlay)	Excluded from OSPF
R2	Tunnel0	10.0.0.2	255.255.255.252	GRE Encrypted Link	Advertised in Area 0

Router 1 (Site 1) Configuration

configure terminal

1. Physical Interfaces

interface GigabitEthernet0/1

description LAN-A

ip address 192.168.10.1 255.255.255.0

no shutdown

exit

```
interface GigabitEthernet0/0
 description WAN-Link
 ip address 203.0.113.1 255.255.255.252
 no shutdown
 exit

 2. Phase 1 & 2 IPsec Configuration
crypto isakmp policy 10
 encr aes 256
 hash sha
 authentication pre-share
 group 14
 exit
crypto isakmp key MY_SECRET_KEY address 203.0.113.2

ip access-list extended GRE-IPSEC
 permit gre host 203.0.113.1 host 203.0.113.2
 exit

crypto ipsec transform-set GRE-TS esp-aes 256 esp-sha-hmac
 mode transport
 exit

crypto map GRE-IPSEC-MAP 10 ipsec-isakmp
 set peer 203.0.113.2
```

```
 set transform-set GRE-TS
 match address GRE-IPSEC
 exit

 3. Apply Crypto Map to WAN
interface GigabitEthernet0/0
 crypto map GRE-IPSEC-MAP
 exit

 4. GRE Tunnel Interface
interface Tunnel0
 description Encrypted-GRE-to-Site2
 ip address 10.0.0.1 255.255.255.252
 tunnel source GigabitEthernet0/0
 tunnel destination 203.0.113.2
 tunnel mode gre ip
 ip mtu 1400
 ip tcp adjust-mss 1360
 no shutdown
 exit

 5. OSPF Configuration
router ospf 1
 router-id 1.1.1.1
  Advertise the LAN into OSPF
 network 192.168.10.0 0.0.0.255 area 0
```

Advertise the Tunnel into OSPF

network 10.0.0.0 0.0.0.3 area 0

Do NOT advertise the WAN (203.0.113.0) to prevent recursive routing

exit

end

write memory

Router 2 (Site 2) Configurationconfigure terminal

1. Physical Interfaces

interface GigabitEthernet0/1

description LAN-B

ip address 192.168.20.1 255.255.255.0

no shutdown

exit

interface GigabitEthernet0/0

description WAN-Link

ip address 203.0.113.2 255.255.255.252

no shutdown

exit

2. Phase 1 & 2 IPsec Configuration

crypto isakmp policy 10

encr aes 256

hash sha

```
 authentication pre-share
 group 14
 exit
crypto isakmp key MY_SECRET_KEY address 203.0.113.1

ip access-list extended GRE-IPSEC
 permit gre host 203.0.113.2 host 203.0.113.1
 exit

crypto ipsec transform-set GRE-TS esp-aes 256 esp-sha-hmac
 mode transport
 exit

crypto map GRE-IPSEC-MAP 10 ipsec-isakmp
 set peer 203.0.113.1
 set transform-set GRE-TS
 match address GRE-IPSEC
 exit

 3. Apply Crypto Map to WAN
interface GigabitEthernet0/0
 crypto map GRE-IPSEC-MAP
 exit

 4. GRE Tunnel Interface
interface Tunnel0
```

```
 description Encrypted-GRE-to-Site1
 ip address 10.0.0.2 255.255.255.252
 tunnel source GigabitEthernet0/0
 tunnel destination 203.0.113.1
 tunnel mode gre ip
 ip mtu 1400
 ip tcp adjust-mss 1360
 no shutdown
 exit

 5. OSPF Configuration
router ospf 1
 router-id 2.2.2.2
  Advertise the LAN into OSPF
 network 192.168.20.0 0.0.0.255 area 0
  Advertise the Tunnel into OSPF
 network 10.0.0.0 0.0.0.3 area 0
 exit
end
write memory
```

Verification command list

General Connectivity & Interfaces

- show ip interface brief: Verifies the up/up operational status and assigned IP addresses of all physical and tunnel interfaces.

- show interfaces tunnel0: Displays detailed information about the GRE tunnel, including source/destination, MTU, encapsulation type, and traffic counters.

- ping [remote-wan-ip]: Tests basic underlay reachability between the physical router WAN interfaces.

- ping [remote-lan-ip] source [local-lan-ip]: Tests end-to-end reachability across the tunnel to simulate host traffic.

Routing & OSPF

- show ip route: Displays the full routing table to confirm that static routes pointing to the tunnel or remote subnets are active.

- show ip route ospf: Filters the routing table to display only the networks learned dynamically via OSPF.

- show ip ospf neighbor: Confirms that OSPF has successfully formed a FULL adjacency with the remote router across the GRE tunnel.

IPsec VPN & Encryption

- show crypto isakmp sa: Verifies the Phase 1 (IKE) tunnel status. A successful negotiation usually displays a state of QM_IDLE.

- show crypto ipsec sa: Verifies the Phase 2 tunnel and actively confirms encryption is working. You must see the pkts encaps and pkts decaps counters increasing as traffic passes.

- show crypto session: Provides a high-level summary of active cryptographic sessions and their current status.

- show crypto map: Displays the active crypto map configurations and confirms they are properly applied to the interfaces.

- show access-lists [acl-name]: Verifies the configuration of the interesting traffic ACL and shows hit counters to confirm if traffic is matching the policy.